

Smart Study Assistant

AI-Powered Study Assistant

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SDAIA ACADEMY





Project Overview

Ask Questions

Get instant AI-powered academic answers

Explain Concepts

Break down difficult topics clearly

Summarize

Condense large study materials

Practice

Generate self-assessment questions



The Problem

Complex Topics

Difficult concepts need clear, accessible explanations

Slow Answers

Students need instant access to reliable information

Content Overload

Large amounts of material are overwhelming to process

Self-Assessment Gap

Practice questions are hard to find and generate

Project Goal

Empower Students

AI-powered explanations for any concept

Instant Answers

No waiting — real-time academic support

Efficient Learning

Summarize and assess in one platform



Core Features



Ask Academic Questions

AI-powered answers instantly



Explain Concepts

Break down complex topics



Summarize Content

Condense large materials



Practice Questions

Self-assessment tools



Bilingual Support

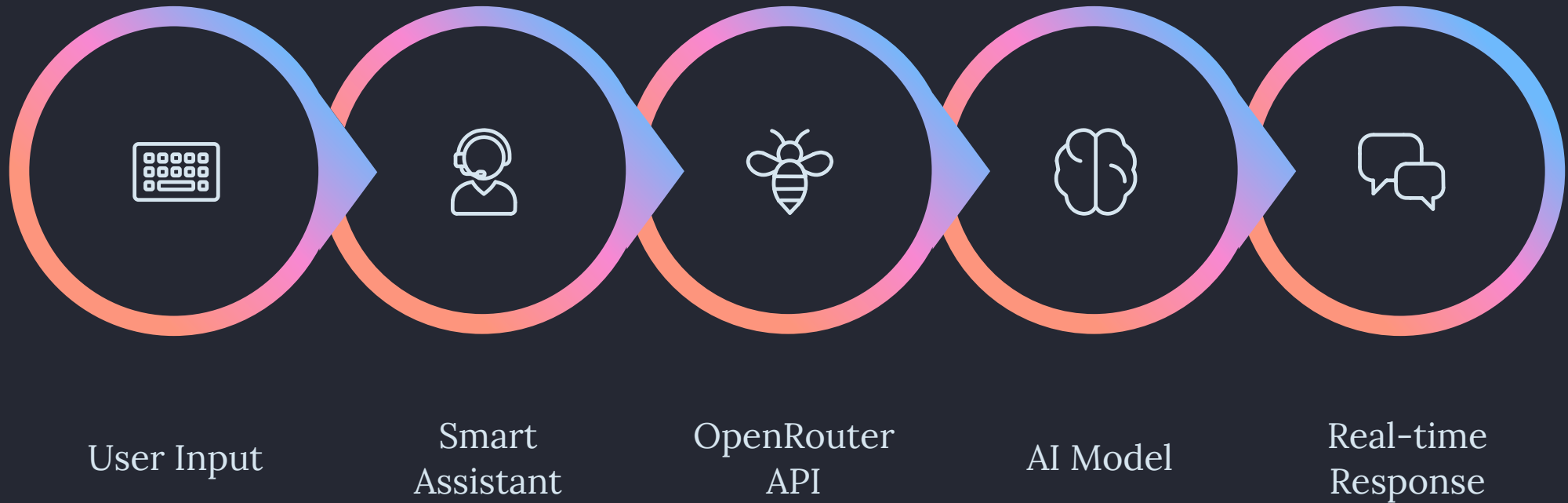
Arabic and English interface



Simple Interface

Easy to use for all students

How It Works



Every question is processed through a secure AI pipeline — from student input to intelligent response in seconds.

Technology Stack

→ Python

Core programming language

→ Streamlit

Web application framework

→ OpenRouter API

AI model integration layer

→ GitHub + Render

Version control and cloud deployment



Project Structure

app.py

Main application and
user interface

translations.py

Arabic and English
language support

requirements.txt

Python dependencies

README.md

Project documentation

SECURITY_RULES.md

Security guidelines and policies

AGENTS.md

AI development instructions

PRD

Project requirements and objectives

Clean, modular architecture for maintainability and scalability.

Security First



No Hardcoded Credentials

API keys never stored in source code



Environment Variables

Sensitive data fully protected



Input Validation

User input sanitized before processing

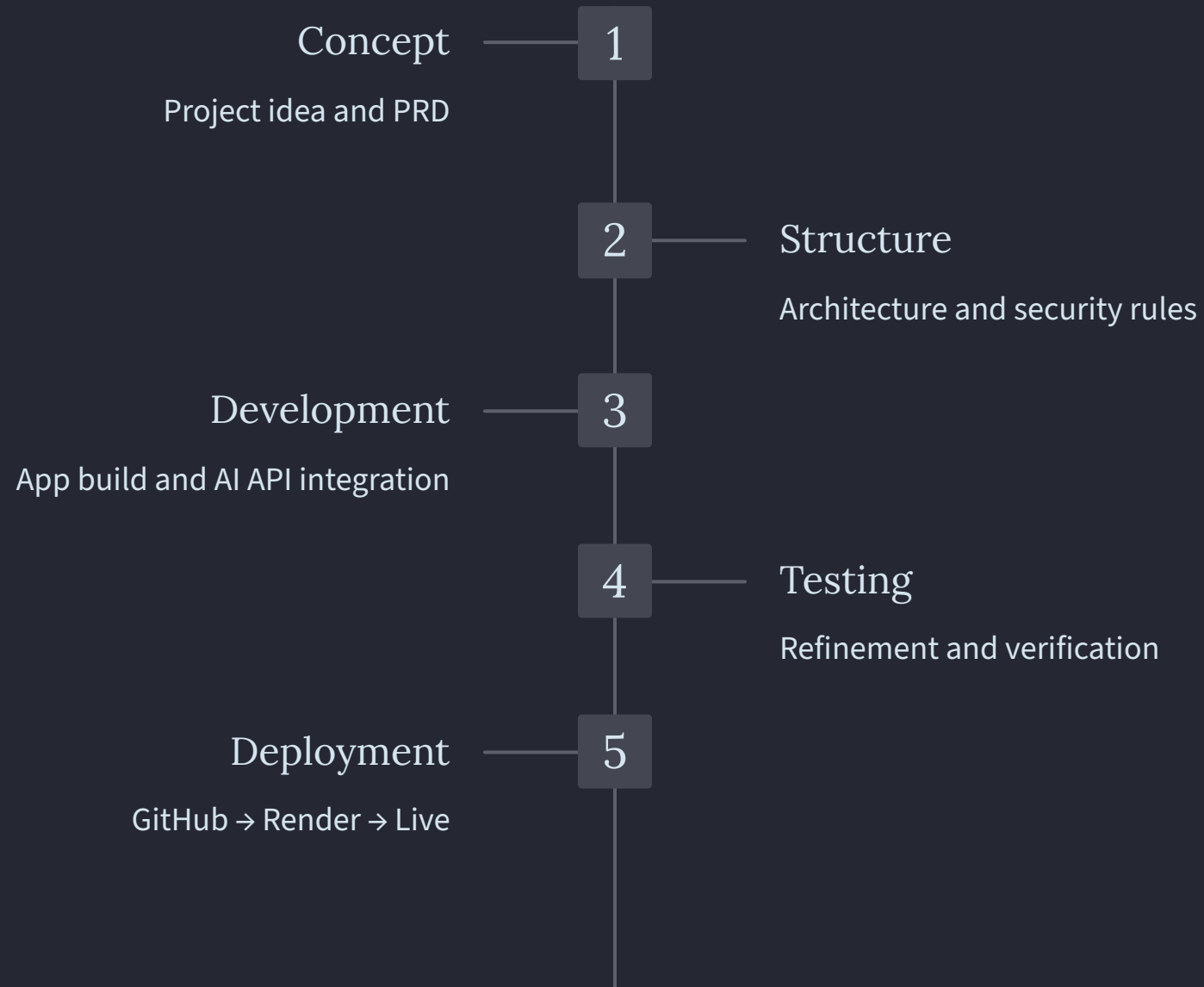


Private Repository

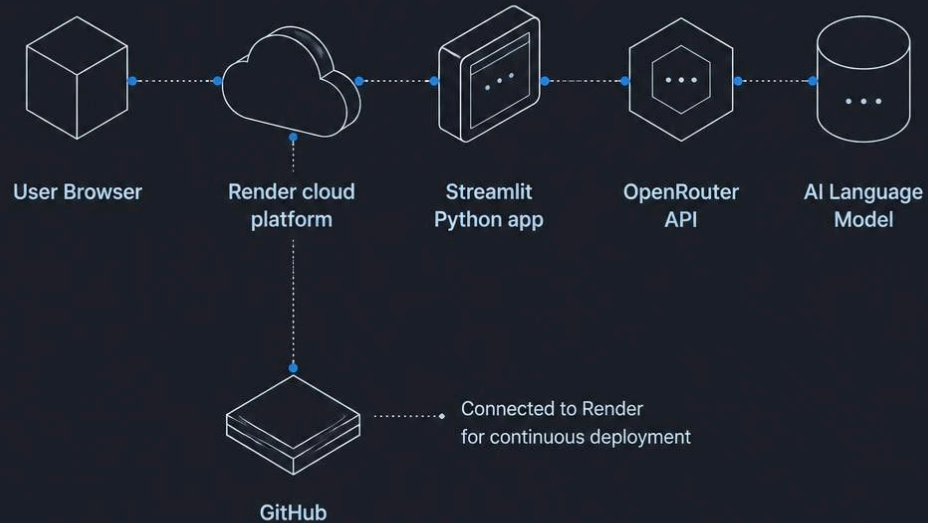
Credentials never exposed on GitHub



Development Timeline



Deployment Architecture



Frontend

Streamlit web interface in the user's browser

Hosting

Render cloud platform for reliable deployment

Backend

Python application processing all requests

AI Integration

OpenRouter API connecting to language models

CI/CD

GitHub connected to Render for continuous deployment